

can suffer from damaging secondary radiation wherein the metal that comprises the habitat's structure creates a scattering effect and/or becomes excited. In contrast, due to their use of non-metallic softgoods as their primary envelope material, inflatables can significantly reduce this dangerous phenomena.

NASA developed the expandable modules being used by Bigelow under its cancelled Transhab program. After elimination of the program, Bigelow Aerospace was granted the sole rights to the expandable module technology. In June 2007, Bigelow Aerospace successfully launched its Genesis 2 prototype space station from a Russian Dnepr rocket. The Genesis spacecraft is an inflatable module that carries 22 cameras, sensors, and avionics to monitor and control the spacecraft. It is 4.4 metres in length and 1.6 metres in diameter but expands to about twice that diameter once in orbit about the Earth. It has an internal volume of 11.5 cubic metres. Its eight solar arrays extend out to 2.5 meters.

The Genesis series of space stations are expandable modules that will be sent into Earth orbit with the intent to provide commercial space habitats for private companies and national space agencies.

The inflatable modules are expected to be more durable and safer than rigid modules because the 1-foot outer skin is made of several layers of a material that is twice as strong as Kevlar. Inner layers are made of other protective materials that add to the safety of the walls. The inflatable walls of the Bigelow space station is expected to be able to withstand micrometeorite and space debris impacts better than rigid walls, say of the International Space Station. The Genesis proved the concept for Bigelow's planned commercial space station, currently called the Space Complex Alpha. It will be made up of the BA 330 module, which can function as an independent space station, or several BA 330 habitats can be connected together in a modular fashion to create an even larger and more capable orbital space complex. Each module – about 330 cubic metres in volume – will contain an independent avionics system to support navigation, re-boost, docking, and other manoeuvring activities. It will also boast of four large windows coated with a film for UV protection, providing an unparalleled opportunity for both celestial and terrestrial viewing. The BA 330 can house up to six persons at a time, and utilizes two propulsion systems on the fore and aft of the spacecraft. The aft propulsion system can be refuelled and reused.

Bigelow Aerospace has currently scheduled the in-orbit assembly of the Space Complex Alpha for 2014. 